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MySQL Implementation Essentials Bootcamp

Backup

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Program Agenda

- Introduction
- Oracle MySQL Enterprise
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- **Backup**
- Replication
- High Availability
- MySQL Enterprise Monitor
- HeatWave

MySQL Backup - Introduction

Why backups?

- Data is your most valuable IP. You have to protect it from loss
- Complete Business Continuity for Disaster Recovery
- Hardware, Software, Developer or DBA error
- Migration and Upgrades (Software and Hardware)
- Create replicas
- Archival (e.g. regulations that requires to keep data for X number of years)
- Move chunks of data
- Setup Test Environments
- And others

Backup and Recovery can be trickier than it appears

- Requires some forethought and planning
  - How do you backup your critical production systems?
  - Backups are resources consuming: when is the best time?
  - Downtime is not acceptable
  - What is the proper backup storage?
- Plan backups monitor and consistency verify
- Plan restore tests
- How integrate with existing
- Document backup policies

Backup types

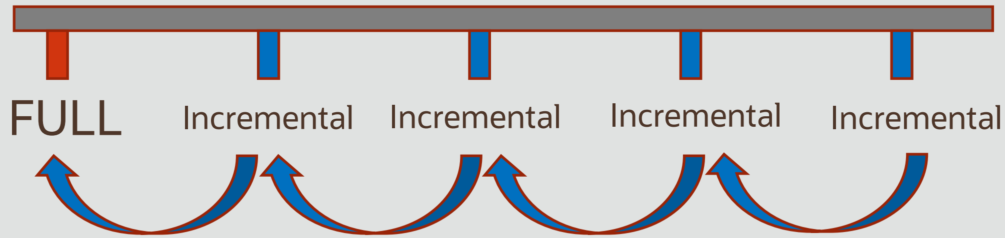
One size does Not fit all “Use Cases” - Pick the right tool(s) for the job(s)

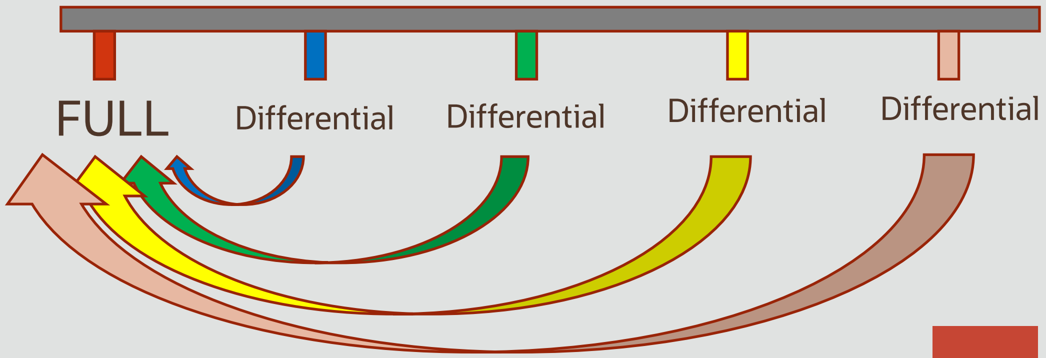
- Logical
  - Convert database and tables to SQL statements
    - portable
    - Server must be running for export/import
  - Generally slower
  - Tools: mysqldump, mysqlpump (5.7 upwards), MySQL Shell, data export
- Physical
  - based on the "raw image" of the DB file(s)
  - DB files must not be changed during the backup
    - Starting DB on physical backups is just like DB restarted from crash event if DB physical backup is taken online (not recommended)
    - Without a specific tool, instance shutdown might be required or lock+flush tables (to allow read but no changes)
  - Tools: MySQL Enterprise Backup, file system copies, Snapshot (LVM, HW, Virtualization)

- Other
  - Activate and save binary logs for PITR
  - Use MySQL Replication for offline backups
  - Use MySQL Replication with replica lag
  - Transportable Tablespaces

Full, Incremental and Differential Backups

- Full Backups
  - A full image (Physical / Logical) of the database
  - Restore a full copy of the DB
- Incremental backups
  - A backup image with the difference from the last point of backup which can be incremental or full backup
  - Restore the last full and in the right sequence all the previous incremental backups (strict dependencies) to the required restore point
- Differential backups
  - A backup image with the difference from the last FULL backup
  - Restore the last full and the Differential that you need





Database Backup Types

Advantages & Disadvantages

Storage Snapshots

- Advantages
    - Quick
    - Feature of Linux LVM, SAN, NAS and virtualized environments
    - Good to use in conjunction with backups
- Disadvantages
    - For consistency it requires a service stop (or put it in read only)
    - It’s a snapshot
        - Still need to make a backup copy – which can be “full” in size
    - Performance may degrade with each concurrent snapshot
        - Snapshots need to be released
    - Cross File System Limitations

mysqldump

- Connect to database and create a SQL file with the commands to rebuild the instance
- Advantages
  - Good for small databases or tables
  - Produce a text file: flexible and portable
- Disadvantages
  - Not always online (requires MVCC or Locks on Tables in the database)
  - Not Consistent (unless transaction is used)
  - Not Incremental (requires a Full Backup every time)
  - slow restore times
- Use mysql client to execute all the commands

https://dev.mysql.com/doc/refman/8.0/en/mysqlpump.html

mysqlpump

- Parallel processing of databases, and of objects within databases, to speed up the dump process (like mysqldump, but parallel execution)
    - databases (queues) and threads
- Not always online (requires MVCC or Locks on Tables in the database, as for mysqldump)
- Dumping of user accounts as account-management statements
- Capability of creating compressed output (using external libraries)
- By default not dump of special internal databases
    - mysql, performance_schema, information_schema, sys
- Not all the mysqldump options are available
- Now deprecated

mysqldump/mysqlpump examples

- Export of all databases for replication, including events and procedures

```
mysqldump […connection parameters…] --all-databases --single-transaction \
    --routines --events --master-data > full_backup.sql
mysqlpump […connection parameters…] --all-databases --single-transaction \
    --routines --events > full_backup.sql
```

- Export users as CREATE USER and GRANT statements

```
mysqlpump […connection parameters…] --set-gtid-purged=off --exclude-databases=% --users > users.sql
```

- Import of the dump

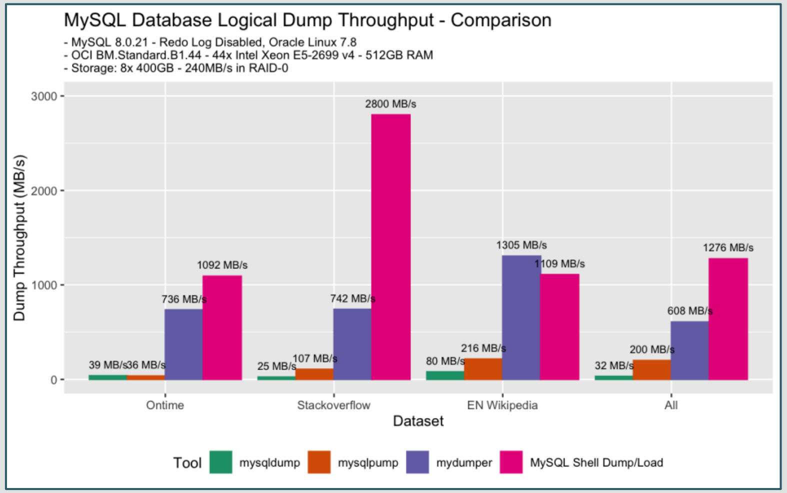
```
mysql -uroot -p -S /tmp/mysql_restore.sock < full_backup.sql
```

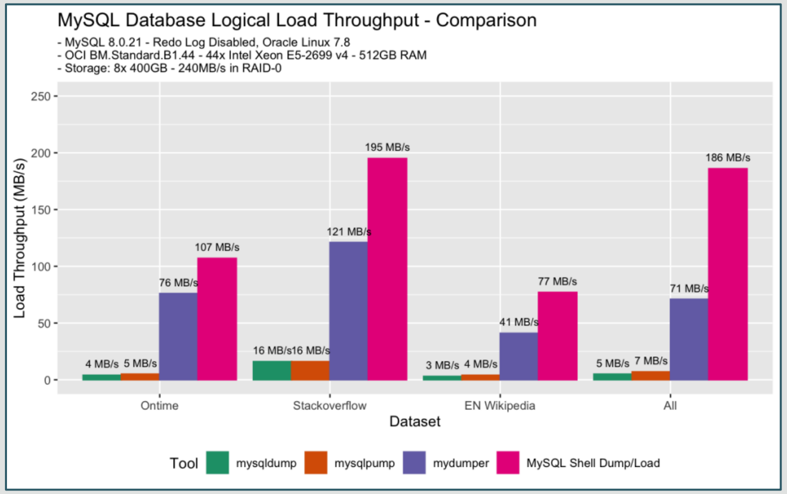
MySQL Workbench: Data Export/Import

- Backed by mysqldump
- Export/Import
  - database(s)
  - database objects/tables
- Export/Import to/from
  - Folder
  - Self contained SQL file

MySQL 8 Shell: Parallel Export/Import

- Supports the dump and import of schemas or tables
  - data streaming from remote storage
  - parallel loading of tables or table chunks
  - progress state tracking
  - resume and reset capability
  - Add primary keys to tables
  - …
- Commands
  - **`util.dumpInstance()`** dump an entire database instance, including users
  - **`util.dumpSchemas()`** dump a set of schemas
  - **`util.loadDump()`** load a dump into a target database





https://dev.mysql.com/blog-archive/mysql-shell-dump-load-part-2-benchmarks/

MySQL Replication

- It's not a backup, but it can improve the backup policy
  - Doesn’t protect from “oops”
- Advantages
  - Rolling “snapshot”
  - Quick Recovery - via failover
  - Non-Blocking
  - Backup overhead not on live production
- Disadvantages
  - Only latest “Point in Time” (point it time keeps moving forward)
  - Not historical
  - Not for archival purposes

MySQL Enterprise Backup

- Advantages
  - Physical Backup so Fast – esp. restores
  - Flexible - many options
  - Archival
  - Scalable
  - Consistent
  - Supported
- Disadvantages
  - Not usable to perform upgrades
  - Not Available from Community Edition

Binary logs

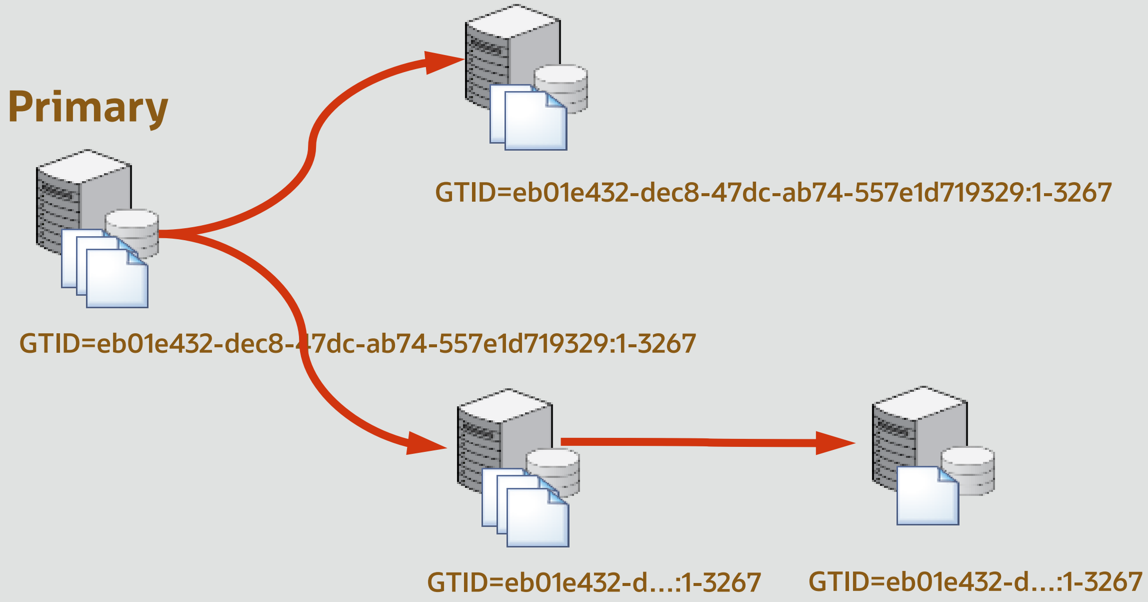
- Set of numbered files containing changes
  - e.g. binlog.000001, binlog.000002, etc.
- Three possible formats
  - statement-based replication (SBR) - propagation of SQL statements
  - row-based replication - logs column changes to individual rows (RECOMMENDED AND DEFAULT in MySQL 8.0)
  - mixed-format replication - SBR is default, but switches to row-based
- Rotate/flush logs with FLUSH BINARY LOGS
  - Closes the logs but does not delete the logs
- Purge can be manual or automatic

```
SHOW BINARY LOGS;
PURGE BINARY LOGS TO ‘mysql-bin.000010';
PURGE BINARY LOGS BEFORE '2019-04-02 22:46:26’;
binlog_expire_logs_auto_purge, expire_log_days, binlog_expire_logs_seconds
```

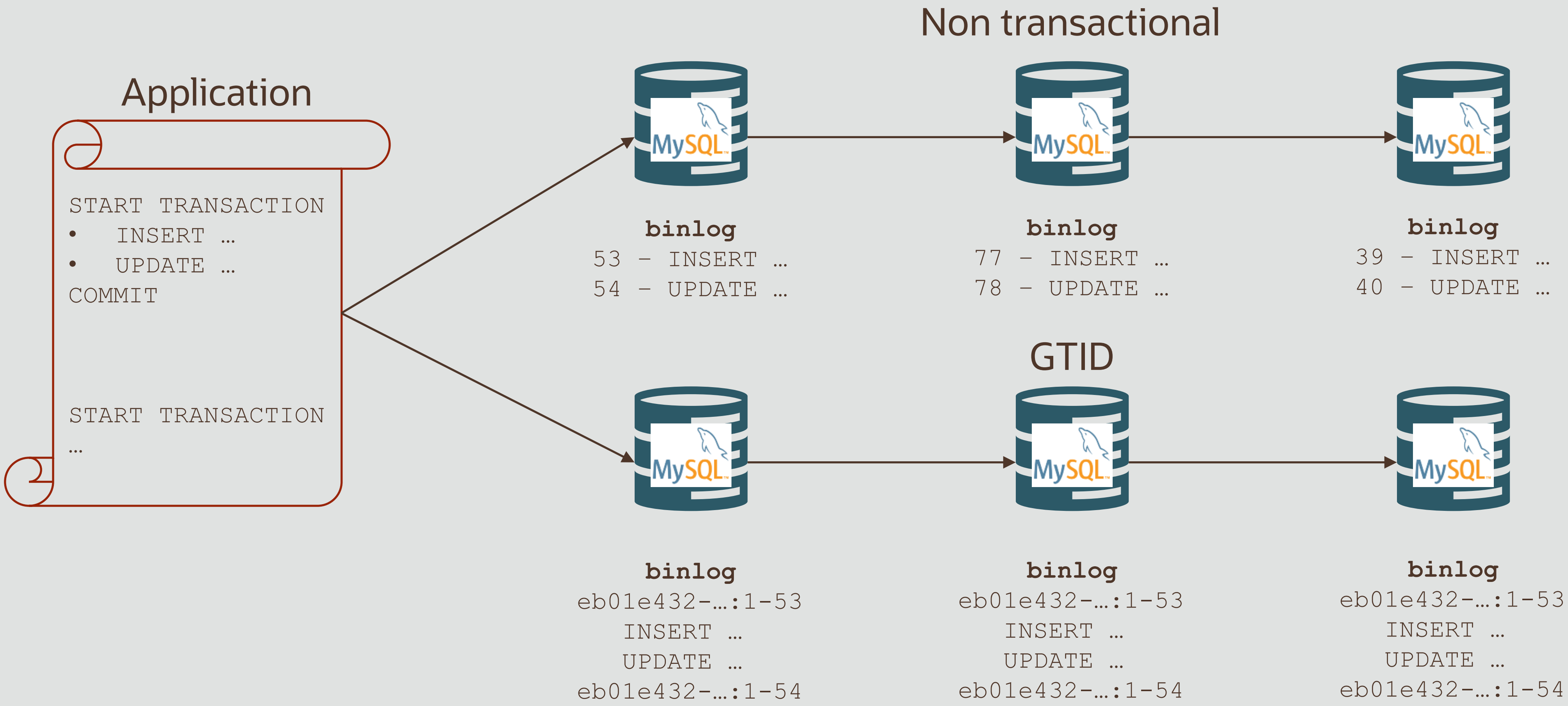
- Can be non-transactional (legacy) or transactional (GTID)

Global Transaction IDentifiers

- GTID: a unique identifier created and associated with each transaction committed on the server of origin (the source)
  - <Server UUID>: <seq. number>
  - The transaction have same GTID in every node involved in replication topology
- Unique across all servers in a given replication topology
- Restriction
  - Updates involving transactional and non transactional storage engines in the same transaction are blocked
  - More restrictions on releases older than 8.0.21



GTID vs. non-transactional binary log



PiTR: process

Restore your database to its state at an arbitrary point

1. Restore last backup
2. Check the actual position (GTID/LSN)

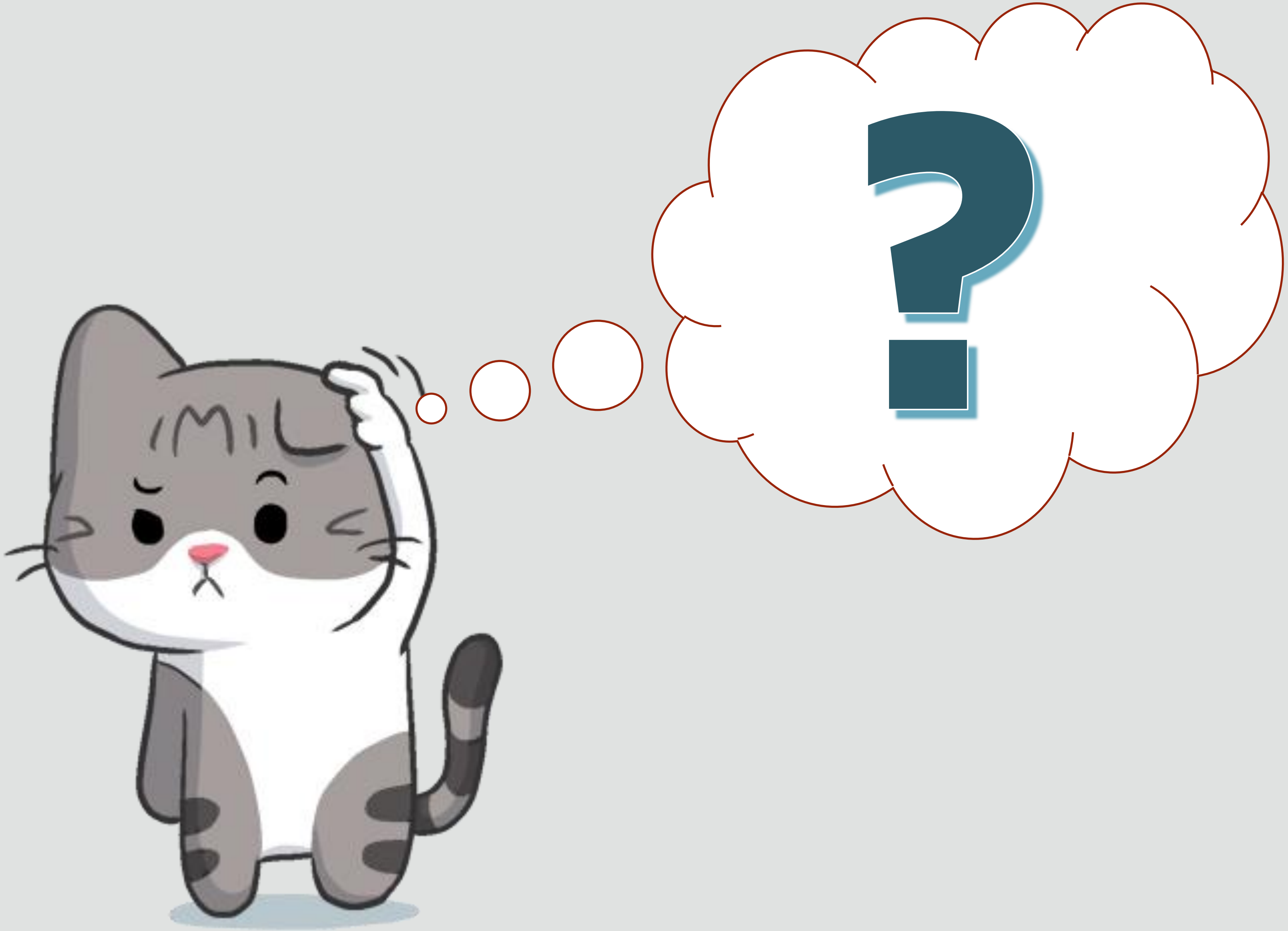
   `mysql> SHOW MASTER STATUS;`
3. Execute events recorded in the binary using mysqlbinlog utility
   - Specify all the binary logs in one command
   - is possible specify when recovery have to start or stop in terms of time or event position

     --start-datetime and --stop-datetime

     --start-position and --stop-position
   - `mysqlbinlog binlog_files | mysql -u root -p`
   - As an alternative to mysqlbinlog, use replication to apply all available binary logs
     - (KB 2009693.1 and KB 2277457.1)

How to save binary logs

- Physical copy of the files on remote system
  - require scripting, to be sure that binary log file is closed when copied
- Create a replica that has copy of the binary logs using mysqlbinlog or replicas
  - mysqlbinlog is not a daemon: monitor it!
  - https://dev.mysql.com/doc/refman/8.0/en/mysqlbinlog-backup.html
  - KB 2180573.1



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5a. MySQL logical backup (mysqldump)

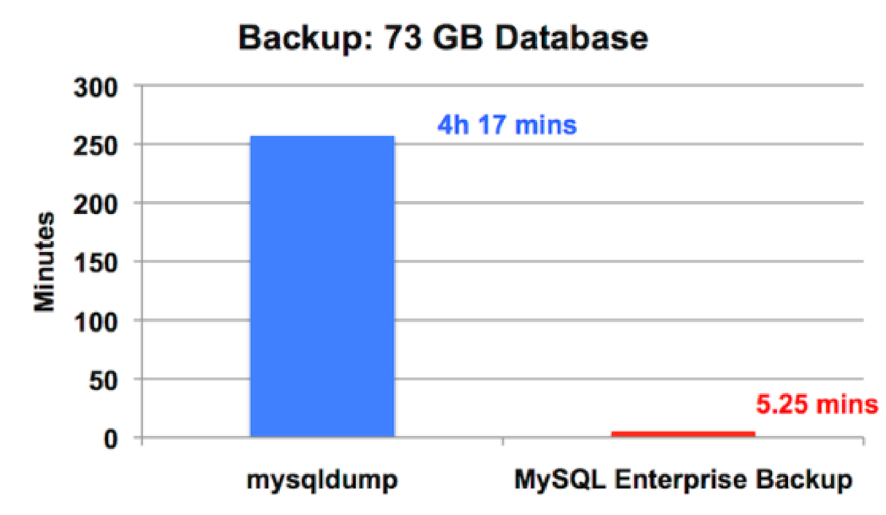
5b. MySQL logical backup (MySQL Shell)

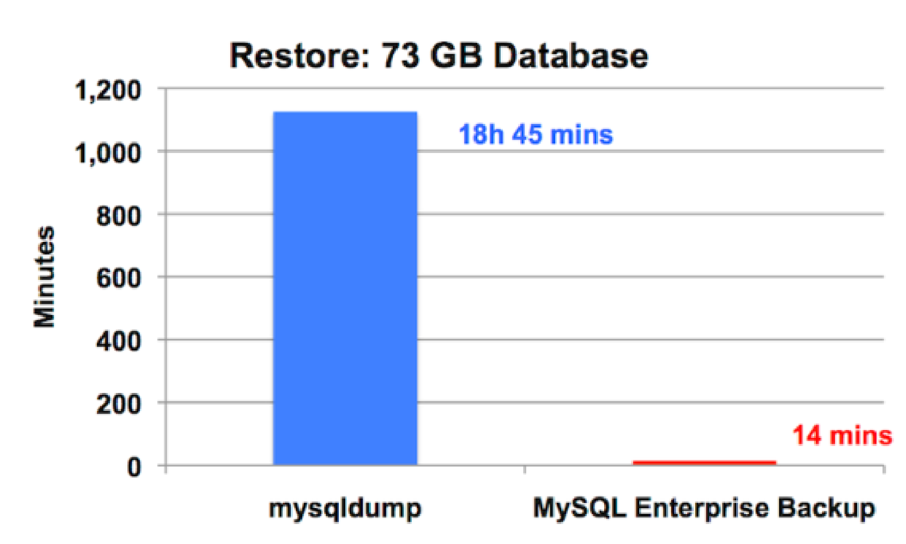
MySQL Enterprise Backup

Features & Benefits

MySQL Enterprise Backup

- Online, non-locking backup and recovery
  - Complete MySQL instance backup (data and config)
  - Partial backup and restore
- Direct Cloud storage backups
  - Oracle Storage Cloud, S3, etc.
- Incremental backups
- Point-in-time recovery
- Advanced compressed and encryption
- Backup to tape (SBT)
- Optimistic backups
- Cross-Platform (Windows, Linux, Unix)



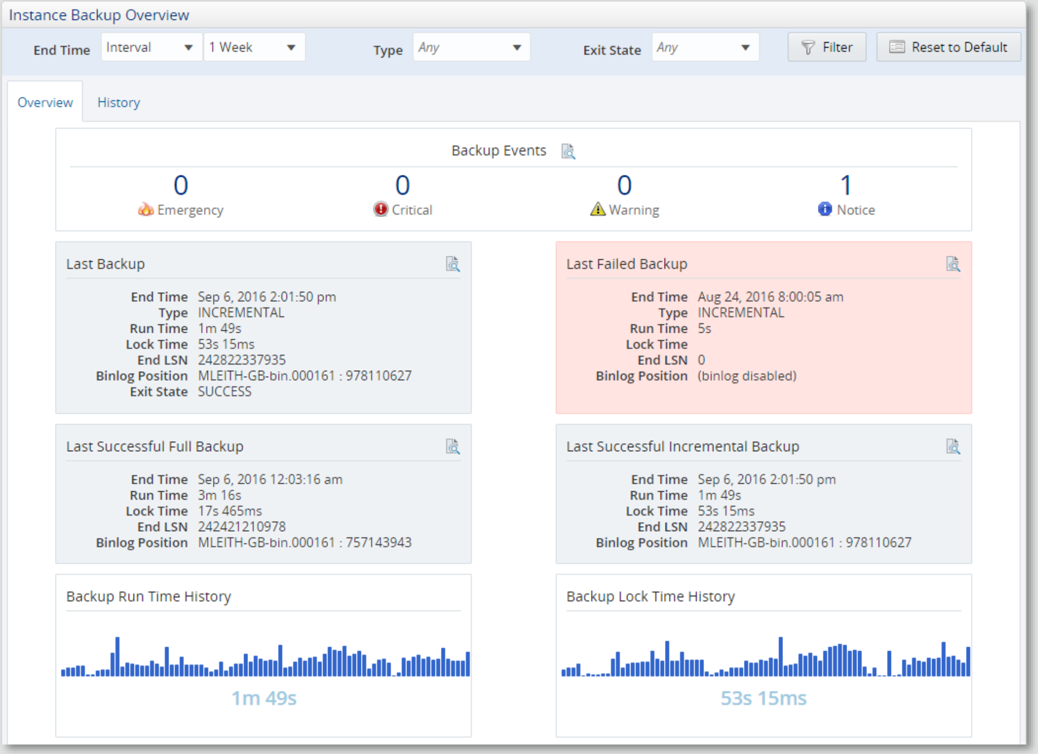


MySQL Enterprise Backup download

- MEB is available from MySQL Enterprise Edition
  - It available in all MySQL Server Enterprise Edition installations and as separate package
  - Download from
    - http://edelivery.oracle.com (trial, only last release)
    - http://support.oracle.com (all the releases)
- MEB 3.12 supports MySQL 5.6, MySQL 5.5
- MEB 4.0.x/4.1.x supports only MySQL 5.7
- MEB 8.0.x supports only MySQL 8.0.x
- MEB of a specific LTS release supports only instances of the same LTS release
  - Major release driven
  - Recommended the last version available of MySQL Enterprise Backup
- MEB of a specific Innovation supports only instances of the same LTS release
  - Major+Minor release driven

Enterprise Monitor integration

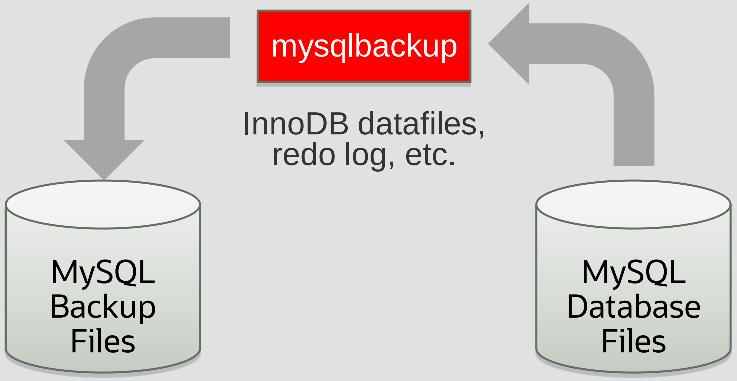
- MySQL Enterprise Monitor provide a Backup Dashboard
- Monitor backup usage and health
  - Across your entire datacenter
- Drill into backup job details
  - Allowing for easy backup recovery
- Supports all backup types
- Alerting on significant events
  - Poor backup performance
  - Backup job failures
  - Out of date backups



Backup Process

Short summary

1. The InnoDB data files, redo log, binary log, and relay log files (except for the log files currently in use) are being copied into the backup, while the database server operates as usual
2. A backup lock is applied on the server instance
   - It blocks DDL operations (except for those on user-created temporary tables), but not DML operations (except for those not captured by the binary log, like administrative changes to the database) on InnoDB tables
3. FLUSH TABLES ... WITH READ LOCK statement is applied on all non-InnoDB tables
4. A brief blocking of logging activities on the server is applied, to collect logging-related information like the current InnoDB LSN, binary log position, GTID, and so on
5. The read lock on the non-InnoDB tables is released
6. Using information from step 4 above, the relevant portion of the binary or relay log file currently in use is copied
7. The backup lock on the server instance is released
8. The redo log files not yet copied before, as well as all the metadata files for the backup, are copied or created
9. The backup operation is completed



https://dev.mysql.com/doc/mysql-enterprise-backup/8.0/en/meb-backup-process.html
https://dev.mysql.com/doc/refman/8.0/en/lock-instance-for-backup.html

MySQL Enterprise Backup files

- MySQL Enterprise Backup save the datadir content
- Raw files are backed up with mysqlbackup
  - InnoDB Data
    - ibdata* files : Tablespace files
    - .ibd files : Per Table data files
    - ib_logfile* files : Log files
  - All other files
    - .MYD / .MYI : MyISAM data and index files
    - others : relay log, binlog, config files, etc…

Restore process

1. Clean destination directory
2. Restore the database using mysqlbackup with the commands copy-back-and-apply-log or copy-back
   - it's important to apply all pending changes to raw copies
3. Depending on how you are going to start the restored server, you might need to adjust some files
   - ownership of the restored data directory
   - my.cnf/mi.ini configuration files
   - mysqld-auto.cnf
   - auto.cnf

Configuration files

- It doesn't save the original one my.cnf/.ini and create copies of configuration file
  - use a standard OS backup for the original my.cnf/my.ini
- It creates configuration files for restore
  - backup-my.cnf: Records the crucial configuration parameters that apply to the backup
  - server-my.cnf: Contains values of the backed-up server's global variables that are set to non-default values
  - server-all.cnf: Contains values of all the global variables of the backed-up server
- Restore rename mysqld-auto.cnf to backup-mysqld-auto.cnf
  - It contains persisted variables
  - Important to restore the original name when you want to restore the original instance or create a replica
- Restore rename auto.cnf to backup-auto.cnf
  - It contains server UUID
  - Important when you want to restore the original instance
  - Don't rename to create a new replica

MySQL Enterprise Backup examples

- Full backup

```
mysqlbackup --defaults-file=/etc/my.cnf --port=3306 --host=127.0.0.1 --protocol=tcp --user=root \
   --backup-dir=/home/mysql/backup/full backup
```

- Restore

```
mysqlbackup --defaults-file=/etc/my.cnf --backup-dir=/home/mysql/backup/full \
   copy-back-and-apply-log
```

- Incremental backup (InnoDB)

```
mysqlbackup --defaults-file=/etc/my.cnf
   --incremental --incremental-base=history:last_backup --backup-dir=/home/dbadmin/temp_dir \
   --backup-image=incremental_image1.bi backup-to-image
```

MySQL Enterprise Backup options

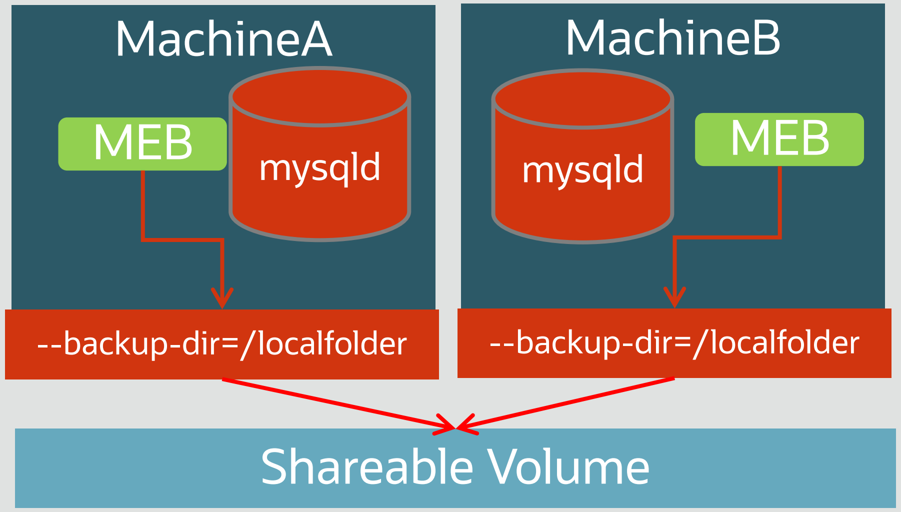
Just some of the many options that can be used

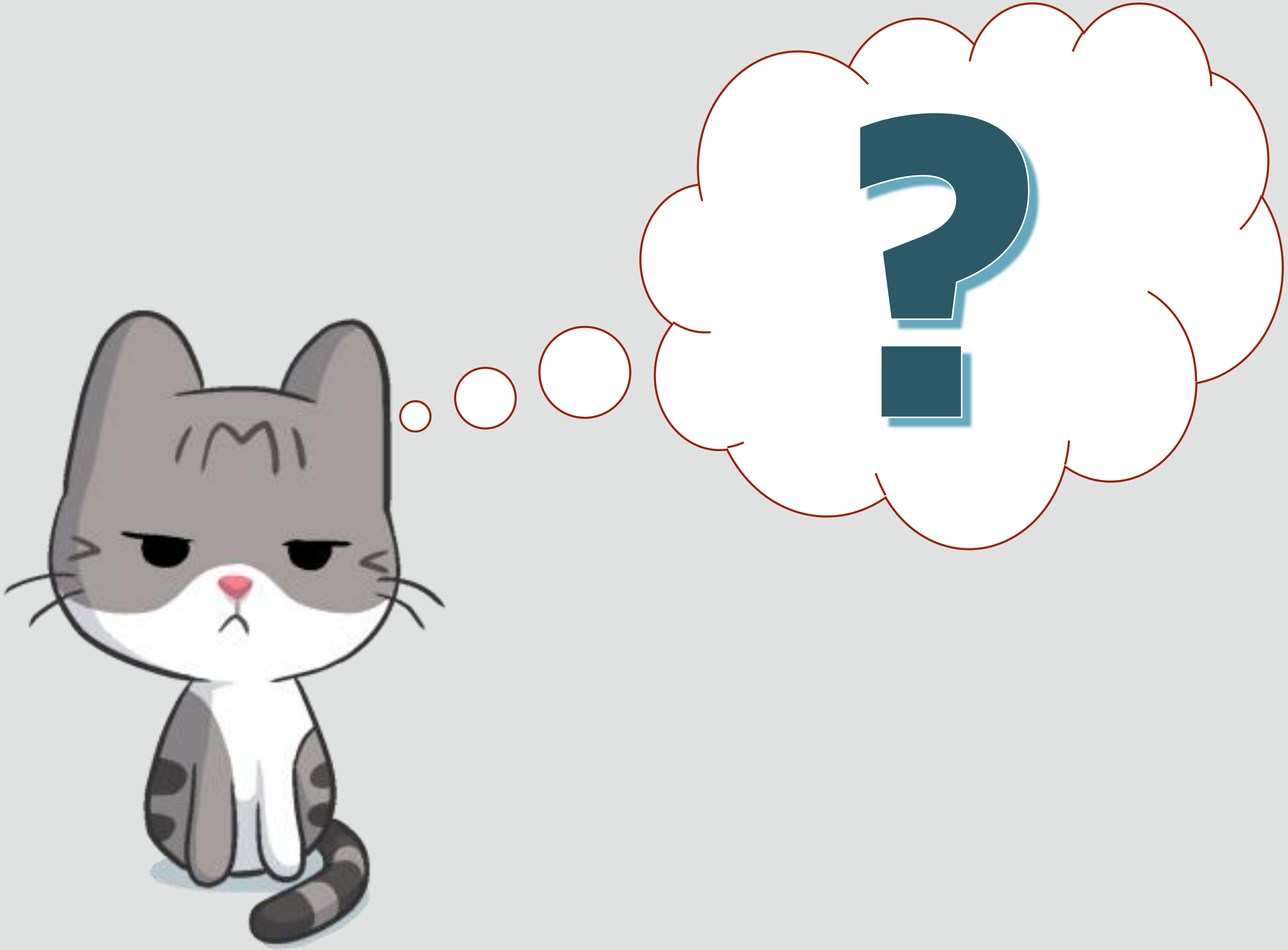
- Encryption
  - --encrypt (--decrypt to restore)
  - --key=<hex key string>
  - --key-file=<keyfile>
  - Generating key-file : e.g. openssl rand 32 -hex > mykeyfile
  - Only valid for Image
- Compression
  - --compress
  - --compress-level=[0 - 9]
  - only data files of the InnoDB format are compressed
- Using the SBT Interface
  - Backup directly to your company's MMS

https://dev.mysql.com/doc/mysql-enterprise-backup/8.0/en/mysqlbackup.tasks.html

Sample MEB Backup Deployment

- Multiple MySQL Instances deployment
  - Each installation with MEB
- Backup to Local Folder / Image File
  - The local folder is specified with - --backup-dir
  - The local folder can be mounted to external NFS or shared volume
- Timestamp
  - option : --with-timestamp (Creating folder with Timestamp), storing multiple copies of backup based on timestamp





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5c. MySQL Enterprise Backup